



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Artificial Intelligence

DS-512-MJ

Interview Preparation — Topic-wise Q&A

Semester I · Elective

Curated interview questions and answers for Artificial Intelligence, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Agents & search

Q. What is an intelligent agent?

An entity that perceives its environment through sensors and acts upon it through actuators to achieve goals.

Q. Informed vs uninformed search?

Uninformed (blind) search (BFS, DFS, UCS) uses no domain knowledge; informed search (Best-First, A*) uses heuristics to guide the search efficiently.

Q. Explain the A* algorithm.

A* selects the node minimising $f(n) = g(n) + h(n)$, where g is cost-so-far and h is a heuristic estimate to the goal. It is optimal if h is admissible.

Q. What is the difference between BFS and DFS?

BFS explores level by level (complete, optimal for unweighted graphs) using a queue; DFS goes deep first using a stack/recursion (memory efficient, not guaranteed shortest path).

Knowledge & reasoning

Q. What is knowledge representation?

Encoding information about the world so an AI system can reason — via logic, semantic networks, frames or production rules.

Q. Propositional vs predicate logic?

Propositional logic deals with true/false statements; predicate (first-order) logic adds objects, predicates and quantifiers (" , ") for richer expressiveness.

Q. Forward vs backward chaining?

Forward chaining reasons from facts to conclusions (data-driven); backward chaining starts from a goal and works back to supporting facts (goal-driven).

